

CyberInsight AI: Intelligent Cyber Threat Analysis Platform

Abstract

Cybercrime is increasing rapidly due to the widespread use of smartphones, computers, and the internet. Users frequently receive malicious APK files, executable files, phishing URLs, and scam QR codes through messaging applications, emails, and social media. Many users unknowingly install or execute these files because existing cybersecurity tools often provide technical reports that are difficult for non-technical users to understand. This creates a need for a simple, user-friendly platform that not only analyzes suspicious files but also explains the associated risks in an understandable manner.

CyberInsight AI is an AI-assisted cybersecurity platform designed to perform **static analysis** of APK files, EXE files, and URLs before they are installed or executed. The system examines file permissions, Android manifest information, digital certificates, metadata, SHA-256 hash values, and URL characteristics to identify potential security risks. Artificial Intelligence is then used to convert these technical findings into easy-to-understand explanations, assign a risk score, and provide security recommendations to the user.

Unlike traditional malware detection tools that primarily display technical results, CyberInsight AI focuses on improving cyber awareness by helping users understand **why** a file or URL may be risky. The proposed system complements existing security solutions by combining multiple static analysis modules with AI-assisted threat explanations in a single platform. In the future, the project will be extended with dynamic malware sandboxing, threat intelligence integration, browser extension support, and machine learning-based malware prediction, making it a more comprehensive cybersecurity solution.